Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/06/2007
PI Name:	Townsend, Lawrence W. Ph.D.		
Project Title:	Radiation Transport Code Development for Space Radiation Shielding Applications		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Knoxville	**State:**	TN
Zip Code:	37996-2300	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	12/01/2002	**End Date:**	02/28/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	8	**No. of Master' Degrees:**	5
No. of Master's Candidates:	8	**No. of Bachelor's Degrees:**	2
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 2/28/07 per PI (4/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pinsky, L. (University of Houston) Empl, A. (University of Houston) Ferrari, A. (CERN & INFN) Cerutti, F. (INFN) Garzelli, M.-V. (INFN) Ranft, J. (CERN & Univ. of Leipzig (Germany)) Gabriel, T. (Scientific Inv. & Develop.) Sihver, L. (Roanoke College/Chalmers University) Adams, J. (NASA MSFC) Watts, J. (NASA MSFC) Wilson, T. (NASA JSC) Moussa, H. (University of Tennessee) Qubain, E. (Roanoke College) Norbury, J. (Worcester Polytechnic Institute/University of Southern Mississippi) Mayes, B. (University of Houston) Andersen, V. (University of Houston) Fasso, A. (SLAC)		

	Vlachoudis, V. (CERN) Smirnov, G. (CERN) Battistoni, G. (INFN) Ottolenghi, A. (INFN) Panini, G. (INFN) Ballarini, F. (INFN) Sala, P. (INFN)
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Performance Goal Text:	
Task Description:	The task, to be carried out by a consortium of researchers from the University of Tennessee (lead institution), the University of Houston, Roanoke College, Worcester Polytechnic Institute/University of Southern Mississippi and various government and national laboratories, involves the development of a standard set of radiation transport computer codes for NASA applications. This effort involves further improvements of the Monte Carlo codes HETC and FLUKA. Included in these improvements is the development of nuclear reaction databases necessary to extend these codes to heavy ion transport. Also planned are analyses of the remaining deficiencies in these codes and in the deterministic code HZETRN, and the determination of what is needed to develop a standard NASA radiation shielding code. Error and sensitivity analyses will be performed to determine errors induced in calculated particle spectra by uncertainties in fragmentation cross sections. The improved codes will be validated by comparisons with measured transport data as they become available. Methods of solving the Boltzmann equation for 3-D radiation transport will also be explored using the numerical method of lines approach. Finally, techniques to facilitate the geometric modeling of complex and detailed 3-D configurations such as spacecraft for use with these transport codes will be explored. To develop a standard space radiation transport code/set of codes for NASA applications. This goal will be accomplished by further development of the HZETRN, HETC and FLUKA radiation transport codes. The ability to simulate the radiation environment in space and the transport of these particles through spacecraft shielding materials is crucial to concerns such as crew health and electronic equipment performance. The need exists to understand both the actual measurement of crew radiation doses recorded on current missions as well as to explore the predicted doses for future missions and new spacecraft designs. These planning tasks call for significant flexibility to simulate a wide variety of conditions such as those likely to exist within spacecraft in interplanetary space, and on the surface of the Moon and Mars. In addition, many experiments need to simulate the effects that space radiation has on their instruments. These effects can either be direct as part of the actual measurements or as a background to it. Further, there can be operational concerns such as degradation of instruments from radiation damage or data interference as in the case of Single Event Upsets. The improvement in the accuracy of the software that NASA currently uses for these tasks that will result from this project will substantially increase the reliability of the projected results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The HETC-HEDS and FLUKA radiation transport codes that were further developed under this project have applications far beyond space radiation protection research and shielding. These codes can, and are utilized, for research in high energy physics, accelerator physics, and radiotherapy applications, and detector development for space applications.
	1. All work on the HZETRN space radiation transport code, except for the development of a Numerical Method of Lines (NMOL) approach to the solution of the 3D Boltzmann equation, was turned over to NASA personnel at NASA Langley Research Center, who were integrated into the Space Radiation Transport Code Development consortium, but were funded under a different instrument, beginning in year 1. All work regarding HZETRN testing and development then became the responsibility of NASA Langley personnel. Work on the Numerical Method of Lines approach to solving the 3D Boltzmann transport equation at Roanoke College has been terminated. The method works well for one-dimensional transport, but does not work well for two-dimensional transport. Based upon the failure of the method in two dimensions, the decision was made to abandon any effort to extend it to three dimensions. 2. The HETC radiation transport code has been modified to enable it to be used to simulate the transport of high-energy heavy ion radiation fields through shield materials and the human body. The new code, called HETC-HEDS, is now capable of carrying out a full 3D transport of the complete space radiation spaectrum. These modifications incorporated in HETC include all physics necessary to transport heavy ions and to describe their nuclear interactions with the target atoms and nuclei. The details of the nuclear interactions between the incident high-energy heavy ions (so-called HZE particles) are provided by a HZE particle event generator, which models the collisions between the transported fields and the target nuclei. The event generator decides what type of collision has occurred (elastic or inelastic) and for inelastic collisions, the secondary radiation particle types, momenta and energies, and directions of travel. Except for mesons produced by heavy ion interactions, all secondary particle types are accounted for by the event generator. Models for pion production in high energy heavy ion collisions have been developed by personnel from the University of Tennessee and University of Southern Mississippi, but have not yet been incorporated into the event generator. Details of the modified version of HETC-HEDS and the HZE particle event generator have been published in several peer-reviewed journal articles, which are listed in the bibliography. Development of the HETC-HEDS event generator formed the basis of a PhD dissertation for one UT student who graduated in 2004. 3. Validation of the HETC-HEDS radiation transport code has been carried out by comparing predictions of fragment spectra and yields, from the nuclear interactions of high-energy heavy ion beams with thick targets, with measured fluence data provided by the NASA-funded Measurements Consortium, using beams at the Brookhaven National Laboratory (AGS and NSRL). Comparisons with secondary neutron yields measured by Japanese researchers have also been carried out. Some issues remain with the secondary neutron spectra, which are in fairly poor agreement with the Japanese measurements. In addition, comparisons of predictions of transported particle yields from HETC-HEDS have been made with predicted yields using the PHITS and GEANT4 codes. The agreement between HETC-HEDS and the PHITS and GEANT4 codes predictions are good (generally within about 25% for most elemental species). These comparisons have been reported at major national and international conferences and have been published in a

Task Progress:

peer-reviewed journal. Additional comparisons with available data provided by the Measurements Consortium continue. Future modifications to the HETC-HEDS code and the HZE particle event generator will be made, if necessary, from these comparisons with measured data. The HETC-HEDS code is currently being used, under the auspices of a separate NASA contract, to provide benchmark data for the HZETRN series of space radaition transport codes under development at NASA Langley Reseaech Center. The HETC-HEDS and FLUKA codes were used as part of a suite of radiation transport codes in the design and characterization of a linear energy transfer (LET) spectrometer under development by a consortium involving Boston U., MIT, University of Tennessee, and Aerospace Corp. for use on the Lunar Reconnaissance Orbiter (LRO) spacecraft, which will be launched in 2008.

4. Comparisons of predicted cross sections obtained from the HETC-HEDS and FLUKA event generators have been made with data obtained by the Measurements consortium. Work on an improved, quantum mechanical nuclear fragmentation model for use in the HETC-HEDS event generator has been undertaken. This effort uses the OPTFRAG knockout-abaltion collision model developed by the PI (Townsend) at NASA LaRC some years ago. The intent is to eventually replace the NUCFRG2 fragmentation model currently used in the event generator with OPTFRAG. A parallel effort to complete development of UBERNSPEC, a code for producing neutron and light ion spectral and angular distributions for use in the event generator, is alo in progress. Finally, a version of the NASA fragmentation code, QMSFRAG, has been recently obtained for eventual use in providing nuclear fragmentation cross sections for the event generator. A UT student, supported by this grant, is working on further development of the UBERNSPEC model and code as her Ph.D. dissertation.

5. At present, the FLUKA code that has resulted from this project has the full capability to simulate the Space Radiation environment over the full range of energies up to those primary particles detected in air showers. As a result of the efforts of the Houston Group in concert with the FLUKA team at CERN and in the INFN Milan Group, FLUKA now embeds two of the most respected heavy ion event generators in existence, RQMD and DPMJET. Both of these event generators have been checked to validate their ability to replicate the available measurements of heavy ion interactions, including those that have been made by our complimentary NASA-sponsored Measurement Consortium, as well as other existing data.

6. The UH Group has long advocated acquiring data in the energy range above 1 GeV/nucleon for several reasons. Not only is the primary Cosmic Ray flux still significant through these higher energies, but there is currently a wide disagreement between existing nucleus-nucleus interaction models in this regime. The decision was made by NASA to pursue measurements at the AGS during the summer of 2005, before the AGS floor itself was to become unavailable for several years. To that end, and because of the lack of time and funding to develop additional detectors, The UH group agreed to employ existing detectors and expertise to enhance the date acquired during those runs. Additional funding was requested to cover the expenses, and in concert with the LBL Measurement Consortium Group, additional stipend money for a graduate student to work on these measurements as his Ph.D. thesis has been supplied to cover those expenses. In addition, the UH group also participated in the running at the HIMAC facility at NIRS in Chiba, Japan in February 2006. Professor Billy Mayes of the Physics Department at UH has joined our effort for these measurements, along with another graduate student, Mr. Andrea Mairani. During the AGS runs data were taken with C, Si and Fe beams of ~3, 5 and 10 GeV/A incident on thin (~ 0.5 interaction length) targets of C, Al, Fe and Cu, as well as a number of thick and compound targets including polyethylene. Analyses of the data taken represents the Ph.D. thesis project for Mr. Elkhayari, and is currently being analyzed. That analysis is separately funded through 2007 and the results of that effort will be made available to the consortium members to guide the needed modifications to the event generators. During the HIMAC runs data were acquired for Si at 800 MeV/A, Fe at 500 Mev/A and Oxygen at 270 MeV/A. Similar targets to those used at the AGS were used. The data taken will be the analyzed by Mr. Mairani and also made available to the consortium members.

6. Significant progress in the primary effort of the UH Group to enhance the models within FLUKA and the user interfaces to the FLUKA code itself have been made under this work. Dr. Neal Zapp has continued to develop his Hamiltonian Molecular Dynamics-based Nucleus-Nucleus Interaction code, which formed the basis of his PhD dissertation, awarded in May 2007, and now has a working version that includes only the internal elastic scattering cross sections for the constituents. He is in the process of exploring the introduction of the inelastic cross sections for the constituent particles. Dr. Kerry Lee completed and defended his Ph.D. thesis utilizing FLUKA to simulate the response of the MARIE experiment. His Ph.D. degree was awarded in May 2006. Dr. Anton Empl, one of the supported postdoctoral researchers, has developed and released via the FLUKA project website a ROOT-based GUI analysis tool to allow visualization of color-coded fluence plots superimposed on line drawings of the input geometry. Work is in progress to expand that capability to allow the plotting of all standard FLUKA scoring capabilities via an entirely new GUI interface based on the use of the GTK graphics software. Significant progress has also been made on the coding of an XML-based GUI front-end tool, known as FLAIR, to enable the user to start and control FLUKA jobs completely from a GUI environment without the need for command-line interactions. Beta versions of that software are currently under test, and there should be a public release of this software in 2007. In July 2006 the long awaited release of the FLUKA source code under a CERN-INFN license agreement occurred, and along with that several improvements to the core FLUKA code were made, the most visible of which from the user's standpoint is the elimination of the need to make an "EGS-like" pre-run to setup the so-called "pemf" file necessary for the implementation of electromagnetic interactions. Now, that initialization activity is seamlessly embedded within the core FLUKA code. NASA users now have full access to the FLUKA Source code via the FLUKA Web-site and under the terms of the general public license agreement. Considerable work was also done to implement and upgrade the treatment of electromagnetic dissociation in Nucleus-Nucleus interactions in FLUKA by Dr. Georgi Smirnov, making use of large volumes of data that are now accessible from Russian sources.

7. Finally, efforts have been underway to not only improve the tools for the input of geometry information into FLUKA, but to develop a broad tool for general use in the conversion of CAD files into Monte Carlo Geometry input files. The Virtual Monte Carlo (VMC) project at CERN is progressing well and there is currently a version of FLUKA being tested in that new environment. The Generic ROOT-based geometry that is included in this package would be a strong candidate for the common Monte Carlo geometry to be the target of such a conversion software code. 8. Copies of all of the released FLUKA software are being supplied as deliverables under the Consortium contract and are furnished under the general CERN/INFN License Agreement. Copies of the HETC-HEDS software and user instructions have been delivered to personnel at NASA Marshall Space Flight Center. This software packege has used by NASA Marshall personnel, in collaboration with the University of Tennessee, to provide benchmarking data for use by NASA Langley Research Center personnel under a separate funding instrument.

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